

Motivation

We want to match the dexterity of blind humans, and see how far we can go without buying new hardware.

Insight: There's a haptic channel in your control loop

- Under PD control, contact perturbs joint-angle tracking. Deviations from the expected motion inject outside information into the body.
- Environmental **contact backdrives the joints**
 - Robot-object interactions have **unique proprioceptive signatures**
 - Well-chosen contact-rich interactions could let us **infer task state!**

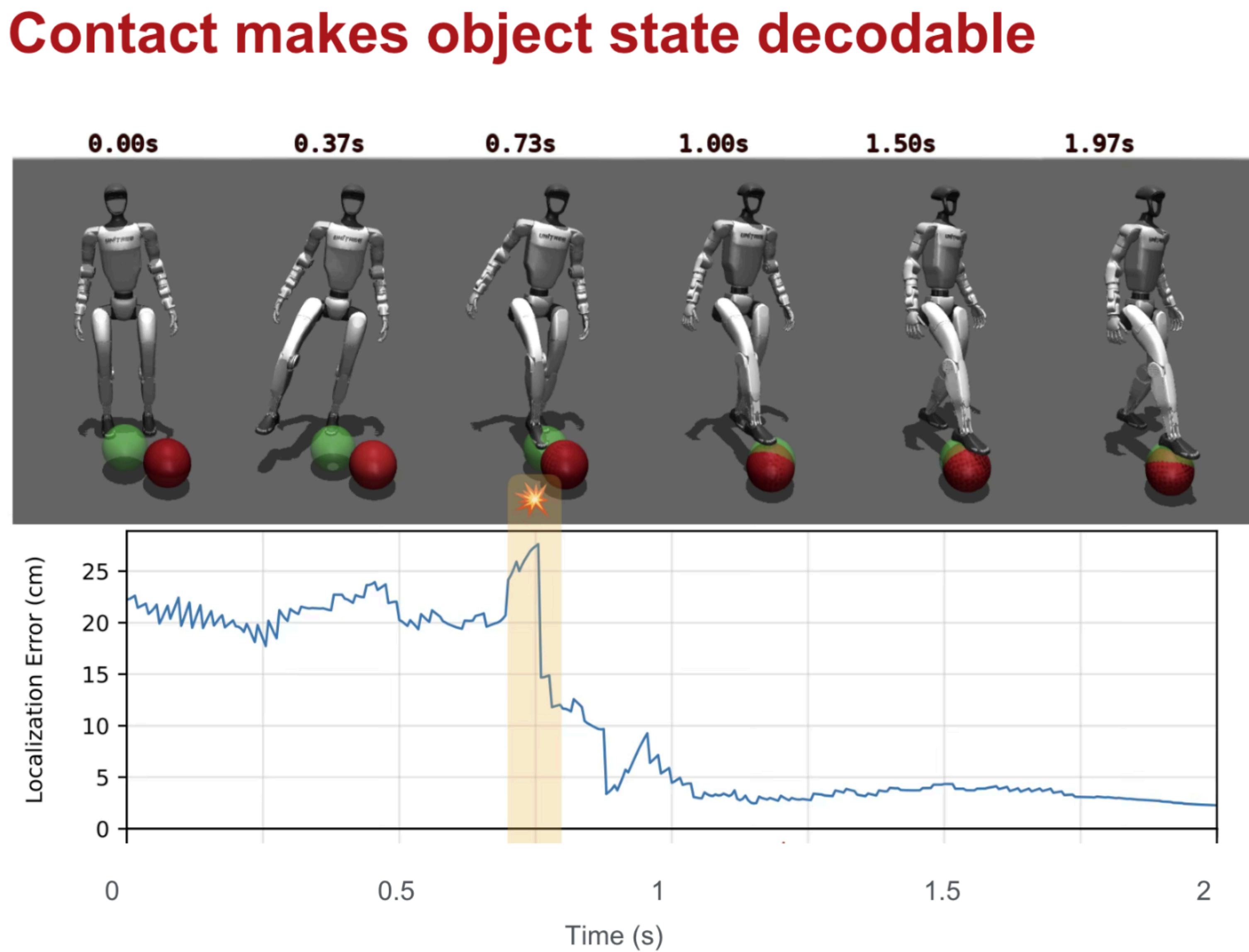
Idea: Train Blind Policies

Train task policies in sim with Deep RL from scratch. **Withhold object state** from the actor. **Randomize object placement** so a fixed reaching motion cannot solve the task.

Inputs: joint states, recent actions, and IMU readings. Just **0.1s** of history for the object-interaction policies.

Learning blind vs. distilling a privileged teacher

Success in simulation	Blind RL	Distilled
Football trapping	92.9 ± 1.2	31.2 ± 33.3
Suitcase lifting	88.3 ± 1.7	34.6 ± 4.1



Had enough of sensormaxxing?
✨Try Interactive Perception. ✨

Blind Whole-body Haptic Dexterity Without Tactile Sensors

 You don't even need an IMU for bipedal walking!

 We kept deleting sensors but the policies still worked

 First skateboard tricks by a robot

 Proprioception is all you need

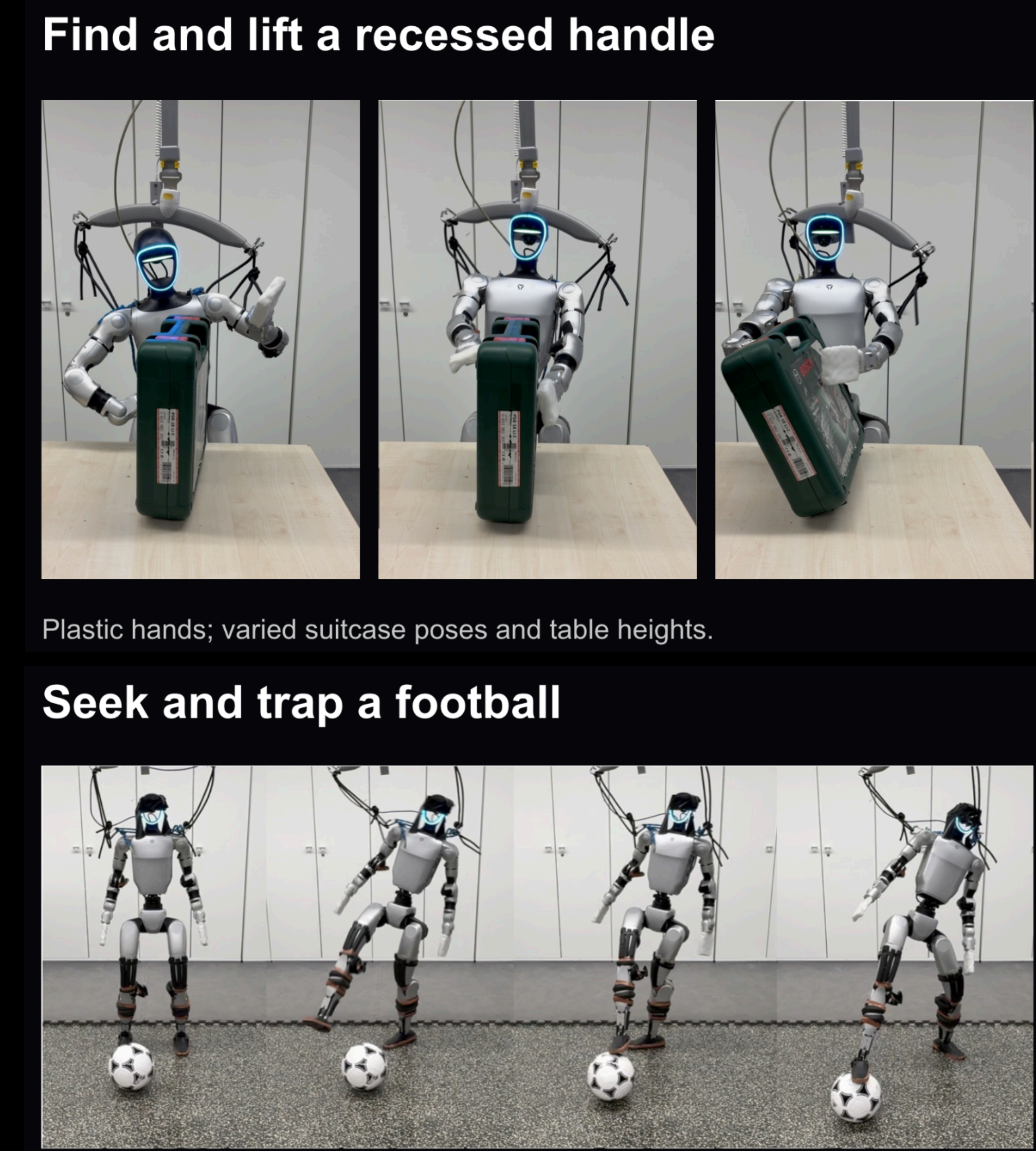
✗ Vision

✗ Mocap

✗ Tactile

✗ Force/Torque

✗ Object pose



We've applied the findings from **Blind Dexterity** to build a complete system for humanoid skateboarding tricks. The skateboard's pose is not perfectly, but still usefully estimateable from **0.1 seconds of proprioceptive history**.



TL;DR Your robot's **joint encoders are a haptic channel**, and leak evidence about the object pose during contact. **Epistemic forcing** encourages policies to **interactively perceive** in the face of sensory deprivation—well enough to pull off precise skateboarding tricks.



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